Business Requirements Document

# Options Strategy Tournament and Agentic Research Application

**Version:** 1.2 — Draft for review

**Prepared:** October 21, 2026 (v1.0)  **Revised:** October 21, 2026 (v1.2)

**Status:** Requirements baseline; unresolved items require owner approval

**Business owner:** User / strategy researcher

Items added or changed in v1.1 are marked **[v1.1]** and in v1.2 **[v1.2]**. Everything unmarked is carried over from v1.0 unchanged.

## 0. Revision history

| Version | Date | Summary |
|---|---|---|
| 1.0 | Oct 21, 2026 | Initial requirements baseline. |
| 1.1 | Oct 21, 2026 | Records decisions from the requirements review: incumbent on Alpaca; SPY-only universe; $10,000 starting equity per account; contract selection driven by the per-trade risk budget; single dedicated Mac mini host (native macOS + Supervisor); remote-IDE development with one repository and after-close changes; startup preflight on a clean tagged commit; data budget under $100/month with the agent on the owner's Claude Pro plan. Adds a proposed technology stack (Section 9.1), deployment and development workflow (Sections 9.2-9.3), new requirements and acceptance tests, the pattern-day-trader rule change, and a status column on the open-decision register. |
| 1.2 | Oct 21, 2026 | Fresh start: the existing (incumbent) strategy is dropped. No incumbent code, history or benchmark is carried forward. Removes Phase 0 and the baseline account; Phase 2 validates two of the five candidates. All other v1.1 decisions stand. |

## 1. Executive summary

Build an application that evaluates five options-scalping strategies concurrently, with each strategy operating in its own isolated paper trading account. The purpose is to determine whether any candidate demonstrates a sufficiently credible, cost-aware trading advantage, and which candidate merits further validation. The system is a controlled research tournament, not a combined portfolio and not an autonomous capital-allocation agent.

The project starts from scratch. No existing strategy, code, trade history or benchmark is carried forward, and no requirement assumes that any of the five proposed strategies is profitable. **[v1.2]**

The five research candidates are VWAP pullback continuation, opening-range breakout, break-and-retest continuation, failed-breakout reversal, and VWAP mean reversion. Their ordering is not a validated profitability ranking. **[v1.2]**

The recommended implementation combines deterministic trading workers with an agentic research and operations assistant. Trading decisions, risk checks, orders, and exits remain governed by explicit versioned code. The agent explains results, investigates incidents, and proposes experiments; it cannot silently change active rules, raise risk limits, or place orders.

The tournament trades SPY options only, starts each paper account with $10,000, and runs entirely on one dedicated Mac mini. **[v1.1]**

Success may mean identifying a candidate for further testing, demonstrating that a candidate's apparent advantage disappears after realistic costs, or concluding that none is ready. A tournament is valuable only if its results are interpretable and its evaluation is reproducible.

## 2. Requirement classification and decision status

Confirmed requirements derive from the conversation. Proposed defaults are implementation recommendations, not approved operating limits. Open decisions must be resolved before the affected capability is enabled.

### 2.1 Confirmed

- Evaluate the five named strategies simultaneously as the target capability.
- Give each strategy a separate paper trading account, independent equity, positions, orders, and risk budget.
- Identify the most profitable candidate without assuming that one must be profitable.
- Preserve fair comparison using common market observations and consistent accounting.
- Include architecture, workflows, risk controls, and research evaluation in the solution scope.
- Do not apply cross-strategy trade suppression intended for a combined portfolio.
- No incumbent strategy: nothing from a prior strategy is reused or used as a benchmark. The no-trade reference (FR-12) is the only baseline. **[v1.2]**
- Broker and market data: Alpaca. **[v1.2]**
- Underlying universe: SPY only. **[v1.1]**
- Starting equity: $10,000 per paper account, set explicitly at account creation (Alpaca's paper default is $100,000). **[v1.1]**
- Contract selection is driven by the per-trade risk budget, not by position size. Order of operations: risk budget -> contract selection -> quantity. **[v1.1]**
- Hosting: all infrastructure runs on one dedicated Mac mini (current model, current macOS) as native macOS processes managed by Supervisor. Nothing else runs on the machine. Remote access is by SSH. **[v1.1]**
- Development: an IDE on a MacBook edits files directly on the Mac mini over SSH. Production code lives in one GitHub repository with a single working directory on the mini. Code changes happen only after the market close, while the trading engine is stopped. **[v1.1]**
- Users: one person (the owner) uses the dashboard, from a laptop. **[v1.1]**
- Budget: under $100 per month for market data and agent usage, excluding hardware. The agent uses the owner's existing Claude Pro plan. **[v1.1]**
- Agent interaction: Telegram is preferred; Claude Code is acceptable when usable remotely from the MacBook. **[v1.1]**

### 2.2 Proposed defaults

- Python-first deterministic trading runtime, PostgreSQL system of record, API/control service, and separate monitoring interface.
- Modular-monolith implementation initially, with adapter boundaries rather than premature distributed services.
- Complete-strategy comparison as the primary experiment; entry-only comparison as a later controlled experiment.
- Two of the five candidates for integration validation, expanding to all five after operational acceptance. **[v1.2]**
- One read-only agentic supervisor rather than five discretionary LLM trading agents.
- Paper-only execution in the initial release; no live routing permissions.
- Equal starting capital and common risk-budget methodology; strategy-specific entry and exit rules.
- Broker: Alpaca paper trading for all tournament accounts, subject to resolving the account-count limit (D02). **[v1.1]**
- Market data: Alpaca Algo Trader Plus (real-time SIP equities and OPRA options). Third-party listings put it at about $99 per month; confirm on Alpaca's pricing page. The free indicative options feed is not acceptable for evaluation. **[v1.1]**
- Technology stack and agent platform as described in Section 9.1. **[v1.1]**

### 2.3 Open decisions

Still open: number of isolated paper accounts obtainable (D02); per-trade risk budget and daily loss limit (D07); expiration and delta policy as a function of the risk budget (D06); exact signal thresholds (D05); order policies (D08); trading session; evaluation duration and statistical confidence policy (D09); cost model (D10); service objectives, backup and retention (D11); and agent tool permissions (D12). **[v1.1]**

Resolved: underlying universe (SPY), hosting (Mac mini), starting equity ($10,000), broker and data (Alpaca), and no incumbent strategy (v1.2). Proposed and awaiting approval: market-data plan and technology stack. See Section 19 for status by decision. **[v1.1]**

# 3. Business problem and objectives

## 3.1 Problem

An untested options-scalping idea cannot show whether its trading hypothesis is weak, its implementation is defective, execution costs overwhelm gross performance, or the evaluation sample is inconclusive. Repeated discretionary changes can obscure this distinction. **[v1.2]** A shared, controlled experiment is needed to compare explicit alternatives without mixing account outcomes or rewriting rules mid-test.

## 3.2 Objectives

**OBJ-01:** Run five independent strategy implementations on a common observation stream.

**OBJ-02:** Measure net profitability, downside, capital usage, and execution sensitivity consistently.

**OBJ-03:** Distinguish strategy failures from data, software, broker, and simulation failures.

**OBJ-04:** Preserve immutable strategy versions and complete experiment lineage.

**OBJ-05:** Support a later confirmation period that was not used to select or tune the candidate.

**OBJ-06:** Reduce research and incident-analysis effort through constrained agent tools.

**OBJ-07:** Support an explicit no-trade/no-promotion conclusion.

## 3.3 Non-objectives

- Guaranteeing profit or discovering an exact best future exit price.
- High-frequency trading, co-location, or ultra-low-latency market making.
- Combining the five strategies into one production portfolio.
- Automatic hedging, cross-account netting, or daily winner rotation.
- LLM-generated discretionary entries in the execution path.
- Online self-modification or automatic promotion to live trading.
- Treating paper results as evidence of live executable returns.

# 4. Stakeholders and responsibilities

**Owner/researcher:** approves strategy specifications, experiment manifests, risk budgets, and promotion decisions.

**Operator:** monitors account health, handles incidents, and executes approved controls.

**Developer:** implements adapters, strategies, tests, reconciliation, and observability.

**Analyst:** evaluates performance and uncertainty without editing historical outcomes.

**Agentic supervisor:** assists through allowlisted research and read-only diagnostic tools.

**Broker/data provider:** external dependency; account availability, data coverage, and simulated execution behavior require verification.

These roles may be performed by one person, but permissions and audit trails should retain the distinction.

In this project the owner performs all human roles. The distinction is kept through separate OS users, database roles, and audit records rather than separate people. **[v1.1]**

# 5. Scope and experiment modes

## 5.1 Primary mode: complete-strategy tournament

Hold capital, evaluation dates, eligible universe, data definitions, cost assumptions, and risk-budget methodology constant. Permit explicit strategy-specific entries, invalidation, targets, and time exits. This answers which complete implementation performed best under the declared constraints.

## 5.2 Secondary mode: entry-signal comparison

Hold contract selection, sizing, and exit treatment as constant as practical; vary entry logic. Setup-specific invalidation differences must be documented because they can affect sizing and outcomes. This answers a narrower question and must not be mixed with complete-strategy rankings.

## 5.3 Execution modes

**Replay:** historical events pass through the same feature and strategy interfaces.

**Shadow:** live observations produce signals and hypothetical decisions without broker orders.

**Paper:** isolated broker paper accounts execute approved intents.

**Live:** excluded from initial release and subject to a separate approval and readiness process.

# 6. Strategy requirements

All strategy descriptions below are research hypotheses. Numeric thresholds, confirmation logic, levels, and indicator definitions must be approved and versioned before testing. No strategy may rely on future information.

## 6.1 Common strategy contract

Each worker receives normalized events and shared timestamped features; owns its setup state; emits structured candidates; and never calls a broker directly. Required outputs include strategy/version, experiment/account association, direction, signal timestamp, snapshot ID, expiration time, trigger and invalidation levels, target policy, time limits, features, and reason codes.

Workers must expose NOT_READY, INELIGIBLE, WATCHING, CANDIDATE, and INVALIDATED outcomes, with more detailed internal states permitted. Session resets, cooldowns, duplicate triggers, re-entry rules, and the handling of existing positions must be explicit. All strategies receive the common event stream; none is forced to trade merely to equalize counts.

All five workers operate on SPY and its option chain. **[v1.1]**

## 6.2 VWAP pullback continuation

**Hypothesis:** a pullback toward VWAP within a directional session can offer a continuation entry after renewed momentum.

**Features:** session VWAP, slope over a declared lookback, price distance, pullback swing, volume, and trend structure.

**Proposed bullish flow:** establish eligible trend; detect pullback; confirm support/structure; break local pullback high; emit bullish candidate. Bearish flow is symmetrical unless explicitly specified otherwise.

**Invalidation:** approved pullback swing/structure breach.

**Exit candidates:** prior extreme, subsequent structure trail, follow-through deadline, stagnation timer, maximum hold.

**Open items:** VWAP calculation, eligible price source, reset policy, slope threshold, pullback tolerance, confirmation, and invalidation buffer.

## 6.3 Opening-range breakout

**Hypothesis:** movement outside an early-session range may continue under specified expansion conditions.

**Features:** session calendar, opening-range high/low, range width, volume/relative volume, price acceptance.

**Flow:** build range; freeze boundaries; detect break; apply confirmation; emit directional candidate.

**Invalidation:** declared return-inside-range rule rather than an undefined discretionary judgment.

**Exit candidates:** target policy, follow-through deadline, trailing policy, maximum hold.

**Open items:** range duration, first permissible entry, breakout buffer, volume normalization, retest requirement, and re-entry count.

### 6.4 Break-and-retest continuation

**Hypothesis:** a broken intraday level that subsequently holds on retest may support continuation.

**Features:** mechanically defined levels, break event, retest depth/duration, confirmation trigger.

**Flow:** define level using available information; observe break; await retest; confirm hold; emit candidate.

**Invalidation:** retest swing or approved structural breach.

**Exit candidates:** next known level, trail, time policies.

**Open items:** eligible level types, level aging, retest tolerance, confirmation rule, and handling of moves with no retest.

### 6.5 Failed-breakout reversal

**Hypothesis:** an excursion beyond a known boundary followed by rejection may favor a move back into the prior range.

**Features:** boundary, excursion magnitude/duration, reclaim/rejection state, range midpoint, VWAP.

**Flow:** define boundary; detect excursion; confirm return inside; confirm failed reclaim where required; emit reversal candidate.

**Invalidation:** failed-breakout extreme plus declared buffer.

**Exit candidates:** range midpoint, VWAP, other approved interior target, time policies.

**Open items:** minimum excursion, acceptance/rejection definition, reclaim test, level source, and trend-regime exclusions.

### 6.6 VWAP mean reversion

**Hypothesis:** an unusually extended price in range-like conditions may revert toward VWAP after reversal confirmation.

**Features:** VWAP, normalized distance, volatility estimate, trend/range classifier, reversal structure.

**Flow:** establish eligibility; detect extension; await reversal confirmation; emit candidate toward VWAP.

**Invalidation:** extension extreme or approved structural threshold.

**Exit candidates:** partial/full VWAP return, time policies.

**Open items:** extension normalization, regime definition, target fraction, confirmation, and treatment of persistent trends.

### 6.7 Existing strategy baseline

Removed in v1.2. There is no incumbent strategy and no sixth benchmark account. New code must not use Alpaca's deprecated pattern-day-trader fields (RK-08). **[v1.2]**

## 7. Fairness and isolation requirements

**FR-01:** Each strategy has an explicit account binding, credential reference, independent ledger, risk state, and order namespace.

**FR-02:** Wrong-account routing fails closed. No fallback to another strategy's account is permitted.

**FR-03:** Identical market observations are broadcast with sequence and snapshot identifiers; workers may decline eligibility independently.

**FR-04:** Opposing or overlapping signals across accounts are allowed. There is no global trade-selection arbiter.

**FR-05:** Common host health controls may pause all affected workers but may not optimize their portfolio exposure collectively.

**FR-06:** Equal starting balances, common risk methodology, and account-level limits are defined in the manifest.

**FR-07:** Risk controls include pending entries and partially filled orders, not just filled positions.

**FR-08:** Strategy-specific exits are permitted only if declared in the comparison mode.

**FR-09:** Configuration changes create a new version and evaluation segment. Historical performance is never overwritten.

**FR-10:** Unequal data outages, account restrictions, or operational incidents are flagged and included in reporting; exclusions require documented rules.

**FR-11:** Account funding/reset events are recorded. Mid-experiment resets cannot erase losses.

**FR-12:** Cash/no-trade is a reference alternative. A relative winner can still fail promotion.

**FR-13:** Every tournament account starts at $10,000, recorded in the manifest at creation. Accounts must not be created at the broker's default balance. **[v1.1]**

# 8. Functional requirements

## 8.1 Market data and shared features

**MD-01:** Ingest underlying prices, eligible option quotes, instrument metadata, session calendar, and broker order/account events.

**MD-02:** Preserve exchange/source time where available, receive time, processing time, sequence, source, and quality flags.

**MD-03:** Deduplicate, detect gaps and out-of-order data, and apply a documented late-event policy.

**MD-04:** Gate trading on quote freshness, required feature readiness, valid bid/ask, and session eligibility.

**MD-05:** Version feature calculations, including VWAP price/volume basis and bar construction.

**MD-06:** Use exchange session calendars and timezone-aware timestamps; store canonical timestamps in UTC and display the selected local timezone.

**MD-07:** Define handling for halts, early closes, session transitions, missing volume, and corporate-action effects.

**MD-08:** Limit option subscriptions to the approved candidate universe without starving required exits of quote coverage.

**MD-09:** Paper and evaluation decisions use real-time OPRA option quotes and SIP equity data. The broker's free indicative options feed may be used only for code testing and must be rejected at startup in paper mode. **[v1.1]**

**MD-10:** Option subscriptions cover SPY only, within the provider's stream symbol limits. **[v1.1]**

**MD-11:** Replay uses the provider's historical options data, available from February 2024 onward at Alpaca. Deeper history is an optional later purchase (D03). **[v1.1]**

## 8.2 Contract selection and sizing

**CS-01:** Select options only after an eligible underlying signal, using current approved contract-selection rules.

**CS-01A:** Contract selection is driven by the account's per-trade risk budget. The pipeline is one-directional: (1) read the per-trade risk budget from the manifest; (2) select the contract whose expiration, delta, spread, and liquidity fit within that budget; (3) compute quantity. Position size never feeds back into contract selection. **[v1.1]**

**CS-02:** Evaluate expiration, right, strike/delta policy, spread, quote age, executable size where available, premium, and liquidity criteria.

**CS-03:** Record rejected contracts and selection reasons, including no suitable contract.

**CS-04:** Calculate planned stop risk and premium-at-risk separately. A stop-based loss estimate is not a guaranteed loss cap.

**CS-05:** Apply common sizing methodology and account-specific available resources. Quantity = per-trade risk budget / per-contract planned risk, rounded down, then validated against quoted size and spread. Rounding and minimum-size rejection must be explicit; a signal whose cheapest eligible contract exceeds the budget is rejected with a reason rather than sized up. **[v1.1]**

**CS-06:** Perform final pre-trade checks after selection because contract price and risk can change.

**CS-07:** Define expiration/session cutoffs to avoid unintended exercise or assignment behavior; broker-specific mechanics require verification.

### 8.3 Account risk

**RK-01:** Enforce per-position, per-underlying, total account exposure, pending-order, daily-loss, and session limits as configured.

**RK-02:** Distinguish an entry halt from an instruction to close positions.

**RK-03:** Reserve risk atomically before creating a submission-ready order intent.

**RK-04:** Recompute reservations after fills, partial fills, cancels, rejects, and reconciliation.

**RK-05:** Permit configurable per-account position count and direction rules; no cross-account suppression.

**RK-06:** Hard risk controls cannot be overridden by an agent-generated score or narrative.

**RK-07:** Per-trade risk budget and daily loss limit are TBD (D07). Both must be set in the manifest before paper execution; activation is blocked while either is unset. **[v1.1]**

**RK-08:** Pattern-day-trader constraints do not apply. FINRA's amended Rule 4210 removed the PDT designation and the $25,000 minimum effective June 4, 2026, and Alpaca implemented the change on that date. The system uses buying_power and must not depend on the deprecated PDT fields, which Alpaca scheduled for removal by July 6, 2026. **[v1.1]**

### 8.4 Orders and positions

**OM-01:** All broker writes pass through an order manager; strategies and the agent have no direct execution path.

**OM-02:** Persist intent and stable client order ID before external submission.

**OM-03:** After uncertain submission, reconcile by broker/client ID before retrying; do not assume a timeout means rejection.

**OM-04:** Handle acknowledgements, rejections, partial fills, cancels, replace requests, late fills, and terminal states.

**OM-05:** Reconcile broker positions, orders, and balances at startup, after disconnects, and periodically.

**OM-06:** Unexpected broker exposure blocks new entries for the affected account until resolved.

**OM-07:** Record intended, submitted, acknowledged, filled, and reconciled quantities separately.

**OM-08:** Protect against duplicate entries and overselling through position-aware serialization.

### 8.5 Exits

**EX-01:** A position manager operates independently of strategy entry evaluation.

**EX-02:** Support underlying invalidation, premium-loss threshold, profit targets, trailing rules, initial follow-through deadlines, no-new-progress timers, maximum holding time, and session cutoffs.

**EX-03:** Each policy states precedence, triggering data, thresholds, and version.

**EX-04:** Exit quantity considers live position, pending exit quantity, partial fills, and broker state.

**EX-05:** Competing triggers cannot create duplicate liquidation orders.

**EX-06:** Data failure policies specify entry blocking and bounded exit behavior; the system must not assume stale quotes are executable.

**EX-07:** Preserve exit reason, realized execution, and any emergency/operator action.

## 9. Architecture and component responsibilities

Market feeds -> normalization -> feature snapshots -> event fan-out -> five strategy workers -> per-account candidate/risk/selection pipeline -> account-bound order manager -> paper broker.

Broker events -> reconciliation -> account/position state -> exit engine -> order manager.

All state changes -> durable event/relational store -> analytics/API/dashboard and constrained agent tools.

Recommended components:

1. Provider adapters: isolate vendor schemas and capabilities.
2. Event normalizer: establishes canonical event contracts and quality checks.
3. Feature engine: calculates common features once per eligible event.
4. Strategy registry: loads explicit versioned implementations.
5. Account runtimes: independent risk, candidate handling, position management, and order state.
6. Order service: centralized implementation with strict per-account partitioning.
7. Reconciliation service: compares internal and broker facts.
8. Experiment manager: owns manifests, lifecycle, versions, and approved changes.
9. Analytics service: calculates consistent metrics and uncertainty.
10. API/dashboard: observes and controls approved operational actions.
11. Agent worker: isolated from timing-critical processing and execution credentials.

Initial deployment should be a modular monolith with bounded asynchronous queues. CPU-heavy research runs outside the trading event loop. Preserve per-account order serialization even if multiple workers are introduced later. Database locks can coordinate local reservations but cannot make broker submission atomic with a database transaction; recovery relies on durable intents, idempotent identifiers, and reconciliation.

Recommended stack: Python, typed validation models, PostgreSQL, FastAPI, and an initially simple Streamlit dashboard. React or another web frontend is optional when UI requirements justify it. These are proposed technology choices rather than procurement decisions.

### 9.1 Proposed technology stack [v1.1]

| Layer | Proposed choice | Notes |
|---|---|---|
| Broker | Alpaca paper trading | Up to 3 paper accounts per login; see D02 for reaching five. |
| Market data | Alpaca Algo Trader Plus (SIP + OPRA) | About $99/month per third-party listings; confirm price. Same data that drives paper fills. |
| Replay history | Alpaca historical options (Feb 2024+) | Optional later: ThetaData Options Value, $40/month, 1-minute options from 2020. |
| Runtime | Python 3.12, uv, alpaca-py, Pydantic, asyncio | Single modular-monolith process for the trading engine. |
| System of record | PostgreSQL (Homebrew) | Orders, ledgers, manifests, lineage. |
| Quote archive and analytics | Daily Parquet files + DuckDB | Keeps raw quote volume out of PostgreSQL. |
| Control API / dashboard | FastAPI + Streamlit | Bound to localhost; reached from the laptop through SSH port forwarding. |
| Secrets | macOS Keychain via keyring | Broker credentials never in files, logs, or agent context. |

| Layer | Proposed choice | Notes |
|---|---|---|
| Process supervision | Supervisor (native macOS) | Engine, API, dashboard, alert bot as separate programs. |
| Ops alerts | Telegram bot in deterministic code | Fires even when the agent is unavailable (AG-09). |
| Agent | Claude Code on the Mac mini, owner's Claude Pro plan | See AG-11. Reached by SSH/tmux, Remote Control, or Telegram channel. |
| Development | IDE Remote-SSH from MacBook; GitHub repo | See Section 9.3. |

Estimated recurring cost: about $99 per month for market data plus the owner's existing Claude Pro plan, within the under-$100 budget for data and agent.

### 9.2 Host deployment on the Mac mini [v1.1]

- One dedicated Mac mini runs every component: trading engine, PostgreSQL, control API, dashboard, alert bot, and agent.
- No inbound ports are exposed beyond SSH with key authentication. Services bind to localhost.
- The agent runs as a separate macOS user with a read-only database role and no access to Keychain items holding broker credentials.
- Power and availability settings: system sleep disabled, automatic restart after power failure, network time sync, and a UPS.
- Nightly pg_dump plus Parquet archive copied off the machine. Restore is tested before paper acceptance (NFR-09).

### 9.3 Development and release workflow [v1.1]

- The owner edits code from an IDE on the MacBook connected to the Mac mini over SSH (Remote-SSH). Files, interpreter, debugger, and git run on the mini; nothing is stored on the laptop.
- Production code lives in one GitHub repository with one working directory on the mini.
- Code changes occur only after the market close while the trading engine is stopped.
- Supervisor starts the trading engine at a fixed pre-market time rather than running it continuously, so an automatic restart cannot load code that is being edited in the evening.
- Startup preflight: the engine refuses to enable paper mode unless the git working tree is clean and HEAD matches the tag named in the approved experiment manifest. It records the commit SHA for every session (NFR-15, AC-21).

## 10. Core workflows

### 10.1 Experiment creation

Owner selects mode, strategy versions, accounts, universe, dates, costs, risk rules, evaluation metrics, and exclusion policy. Validate unique account bindings and provider capabilities. Save immutable manifest and approval. Unresolved mandatory values prevent activation.

The manifest names the git tag for the strategy code, the $10,000 starting balance per account, and the per-trade risk budget and daily loss limit. [v1.1]

### 10.2 Startup and recovery

Run the preflight check (clean working tree, HEAD at the manifest tag, real-time OPRA feed configured); load approved manifest; authenticate paper-only endpoints; verify account identity and mode; reconcile broker state; load/resume strategy and position state; warm required features; check data quality and clocks; enable entries only when readiness passes. Existing positions require management before new signals are accepted. [v1.1]

### 10.3 Entry lifecycle

Normalize event -> update features -> broadcast snapshot -> strategy state transition -> candidate -> eligibility/deduplication -> account risk screening -> read per-trade risk budget -> contract selection within budget -> quantity sizing -> final checks -> reserve risk and persist intent -> submit -> acknowledge/reconcile -> process fills -> activate position exit policy. **[v1.1]**

A candidate has an expiration time. Delayed processing does not authorize entry on an obsolete setup.

### 10.4 Exit lifecycle

Market/timer/broker event -> evaluate active position policy -> choose trigger under declared precedence -> compute safe remaining quantity -> persist exit intent -> submit/reconcile -> update position and risk -> record outcome.

### 10.5 Incident lifecycle

Detect anomaly -> classify affected feed/account/component -> pause relevant entries -> retain position monitoring -> alert operator -> reconcile -> record remedy -> resume only after readiness passes. Global emergency actions and account-only actions remain distinguishable.

### 10.6 Session close

Apply cutoff policy -> cancel stale entry orders -> process configured liquidation -> reconcile open exposure -> mark unresolved incidents -> create equity snapshots and daily report. Never report the account as flat solely because local state says so.

After reconciliation, Supervisor stops the trading engine. The after-close window is the only period for code changes. **[v1.1]**

### 10.7 Research change

Agent or analyst proposes hypothesis -> owner reviews -> run replay -> create new configuration/version -> assign new experiment segment -> approve activation. Active tournament rules are not silently edited.

## 11. Data model and lineage

Required entities:

- experiments and immutable manifests;
- strategy definitions, code/configuration versions, and feature versions;
- account bindings and non-secret provider references;
- instruments, contracts, calendars, and metadata snapshots;
- raw/normalized market events or sufficient approved replay artifacts;
- feature snapshots and strategy state transitions;
- candidates, eligibility decisions, rejection reasons, and contract evaluations;
- risk reservations and limit events;
- order intents, broker orders, fills, cancellations, and reconciliation events;
- positions, lot attribution, realized/unrealized P&L, and equity snapshots;
- modeled execution scenarios and evaluation-ledger entries;
- incidents, operator actions, approvals, and agent tool traces;
- metric definitions, calculation versions, and generated reports.

Every trading record carries experiment ID, strategy/version, account ID, event time, correlation ID, and relevant source identifiers. Corrections append auditable records rather than silently rewriting history. Secrets are stored outside ordinary research tables. Retention and raw quote volume are open capacity decisions.

Each engine session also records the git commit SHA and manifest tag it ran under. **[v1.1]**

## 12. Performance accounting and evaluation

### 12.1 Primary objective

Rank net return on equal starting capital during the common evaluation period, subject to predeclared risk constraints. Rank eligibility and promotion eligibility are separate. An account can be first and still fail every promotion gate.

### 12.2 Metrics

Report net P&L, return, gross P&L, commissions/fees, execution-cost estimates, maximum drawdown, daily returns, average net trade outcome, win rate, payoff ratio, trade count, active days, worst day, largest loss, holding duration, exposure time, premium deployed, turnover, and operational incident counts.

Profit factor and Sharpe-like measures may be included with clear definitions and undefined-value handling. Do not use win rate alone. Explain all denominators, time periods, annualization choices, and treatment of open positions.

### 12.3 Two ledgers

Broker ledger records actual simulated fills, cash movements, positions, and provider-reported results. Evaluation ledger applies consistent cost and fill assumptions to the available observations. Keep the ledgers distinct and reconcile differences.

Do not double-count bid/ask costs already reflected in prices. Midpoint marks may be shown for reference, but report a declared executable-side liquidation estimate where quotes support it. Missing, stale, or crossed quotes create quality flags, not invented precise values.

For passive orders, a quote touching the price is not sufficient proof of a fill in an internal execution model. Record queue/fill uncertainty. Model scenarios consistently across competitors and label modeled outcomes as hypothetical.

### 12.4 Scientific controls

Freeze versions during evaluation. Maintain the history of all attempted variants. Distinguish development data, tournament selection data, and later confirmation data. Evaluate paired daily outcomes on common eligible dates, while documenting each account's operational availability.

Potential analysis methods include block-based resampling to assess dependence and uncertainty, execution-cost stress scenarios, and regime-stratified reports. The exact method, block policy, minimum evidence, and decision thresholds require approval. No fixed trade count guarantees sufficient evidence.

Report whether the leading strategy's margin is uncertain. Avoid declaring a durable winner from a short leaderboard or repeatedly inspected significance tests. If continuous monitoring is used for formal decisions, predefine the stopping/decision policy.

### 12.5 Promotion gates

Proposed gates: positive net performance; acceptable drawdown; adequate evidence; no unresolved material accounting incidents; tolerable conservative-cost performance; and unchanged confirmation-period results meeting approved criteria. These are configurable proposals, not approved thresholds or promises.

The no-trade benchmark has zero strategy trading P&L under the declared cash-return assumption. Infrastructure and subscription costs should be shown separately and, if included in net business profitability, allocated by an explicit policy.

## 13. Agentic capabilities and boundaries

**AG-01:** Explain accepted/rejected signals using recorded evidence.

**AG-02:** Compare strategy results and identify concentration, cost sensitivity, and incident effects.

**AG-03:** Investigate stale feeds, reconciliation mismatches, rejects, and unusual fills through read-only tools.

**AG-04:** Run owner-approved bounded replay/analysis jobs in an isolated environment.

**AG-05:** Propose new hypotheses/configurations with supporting evidence and limitations.

**AG-06:** Produce daily and experiment-end narratives linked to calculation versions and records.

**AG-07:** Do not place orders, change live/paper limits, modify active strategy code, access broker secrets, or promote strategies.

**AG-08:** Treat vendor text and retrieved artifacts as data, not instructions. Log tool calls and restrict filesystem/network access.

**AG-09:** Agent failure or latency must not interrupt data ingestion, risk controls, orders, exits, or reconciliation.

**AG-10:** Agent conclusions cannot replace deterministic accounting or approved statistical calculations.

**AG-11:** Proposed implementation: Claude Code on the Mac mini under the owner's Claude Pro plan, running as a separate macOS user with a read-only database role and no access to broker credentials, so AG-07 is enforced by the operating system rather than by instructions alone. The owner reaches it by SSH (inside tmux), by Remote Control from claude.ai/code or the Claude app, or through the Telegram channel plugin, which is a research-preview feature. **[v1.1]**

**AG-12:** The Telegram channel allowlist contains only the owner. Any allowlisted sender can approve the agent's tool use, so no other account may be added. **[v1.1]**

**AG-13:** Daily narratives (AG-06) run as a headless agent job started by Supervisor after the close. Heavy replay and analysis run as Python jobs, not as model calls, to stay within Claude Pro usage limits. **[v1.1]**

## 14. Dashboard and controls

**Overview:** experiment state, five account health indicators, equity/drawdown curves, net rankings, and no-trade reference.

**Strategy view:** state machine, eligibility, candidates, rejection reasons, positions, active exit policies, and versions.

**Execution view:** quotes, intents, acknowledgements, fills, latencies, partial fills, cancels, and broker/evaluation differences.

**Research view:** daily/regime breakdowns, uncertainty, cost scenarios, variant history, and confirmation status.

**Operations view:** feed lag, queue depth, reconnects, reconciliation status, incidents, and audit log.

Controls must clearly distinguish pause new entries, resume after validation, cancel pending entries, close a position, flatten one account, and global emergency action. Destructive or exposure-changing controls require explicit authenticated confirmation and durable audit records. A dashboard restart must not restart or stop the trading engine implicitly.

The dashboard serves a single user and is reached from the owner's laptop through an SSH tunnel. **[v1.1]**

## 15. Nonfunctional, security, and reliability requirements

**NFR-01:** Deterministic replay uses controlled event order, configuration, clocks, and randomness where applicable; document unavoidable external differences.

**NFR-02:** Establish measured latency budgets for feed processing, candidate evaluation, submission, broker acknowledgement, and exit handling before paper acceptance. No numeric SLA is assumed.

**NFR-03:** Bounded queues expose backlog and fail closed for stale entries rather than silently dropping critical order events.

**NFR-04:** Preserve durable order intents and fills across restart; reconcile externally before resuming.

**NFR-05:** Isolate research CPU/memory use from trading workloads.

**NFR-06:** Use paper-only credentials/endpoints and explicitly reject live mode in the initial release.

**NFR-07:** Encrypt secrets in an approved secret store; redact credentials and sensitive account data from logs and agent context.

**NFR-08:** Restrict control permissions by role; authenticate API/UI access.

**NFR-09:** Maintain backups and test restore; define recovery point/time objectives before deployment acceptance.

**NFR-10:** Monitor clocks, feed health, account identity, provider throttling, and disk/database capacity.

**NFR-11:** Version database migrations, calculations, configurations, and deployment artifacts.

**NFR-12:** Verify licenses, entitlements, and redistribution restrictions for market data and reports.

**NFR-13:** Host resilience: the Mac mini has sleep disabled, restarts automatically after power loss, syncs time over the network, and runs on a UPS. It is a single point of failure; restart recovery relies on NFR-04 and OM-05. **[v1.1]**

**NFR-14:** Network exposure: only SSH with key authentication is reachable; API, dashboard, and database bind to localhost. **[v1.1]**

**NFR-15:** Release integrity: paper mode starts only from a clean working tree whose HEAD matches the manifest tag, and each session logs its commit SHA. **[v1.1]**

## 16. Testing and acceptance criteria

**AC-01 Account isolation:** simultaneous opposing candidates route only to their assigned paper accounts; wrong binding is rejected.

**AC-02 Shared observations:** all five workers reference the same input snapshot for a broadcast event, with independent eligibility outcomes.

**AC-03 Deterministic replay:** repeated controlled replays produce matching candidates and state transitions under the declared environment.

**AC-04 Risk reservation:** concurrent candidates cannot exceed an account's configured limit through a race condition.

**AC-05 Submission timeout:** an accepted-but-unacknowledged order is reconciled without a duplicate entry.

**AC-06 Partial fills:** reservations and exits reflect actual remaining quantities.

**AC-07 Exit race:** simultaneous target/time/invalidation triggers cannot oversell the position.

**AC-08 Stale feed:** entries halt according to policy and an incident is recorded; position handling follows the approved contingency.

**AC-09 Restart:** outstanding broker orders/positions are discovered and managed before new entries.

**AC-10 Versioning:** any parameter change creates a new immutable version/segment without rewriting prior results.

**AC-11 Accounting:** synthetic fixtures reconcile cash, fills, fees, open positions, equity, and drawdown to independently checked expected values.

**AC-12 Comparison:** an unprofitable relative leader is displayed as leading but not promotion-qualified.

**AC-13 Agent permissions:** attempts to submit orders, reveal credentials, or modify active settings are denied and logged.

**AC-14 Agent independence:** trading continues during agent unavailability.

**AC-15 Paper-only:** live endpoint/credential configurations fail startup validation.

**AC-16 Controls:** pause, cancel, flatten, and resume behaviors are distinct and auditable.

**AC-17 Incident fairness:** outages remain visible in evaluation; exclusions cannot silently remove losing periods.

**AC-18 Expiration cutoff:** affected contracts are blocked/managed according to the approved policy.

**AC-19 Scalability:** all five workers run within measured resource/latency budgets for the approved universe.

**AC-20 Recovery:** backup restore and reconciliation tests meet approved recovery objectives.

**AC-21 Release preflight:** a dirty working tree, or a HEAD that does not match the manifest tag, prevents paper mode and is logged. **[v1.1]**

**AC-22 Risk-budget selection:** no selected contract and quantity exceeds the per-trade risk budget; a signal with no budget-feasible contract is rejected with a recorded reason. [v1.1]

**AC-23 Feed guard:** starting paper mode with the indicative options feed fails startup validation. [v1.1]

**AC-24 Agent OS isolation:** the agent's macOS user cannot read broker credentials or write to the trading database. [v1.1]

## 17. Delivery phases and exit gates

**Phase 0 — Audit incumbent:** removed in v1.2; there is no incumbent strategy. [v1.2]

**Phase 1 — Specification/replay:** approve strategy definitions and schemas, build shared features and replay tests. Gate: no look-ahead and traceable decisions.

**Phase 2 — Two-account validation:** two of the five candidates in shadow/paper as approved [v1.2]. Gate: isolation, risk, execution, and accounting acceptance.

**Phase 3 — Five-strategy tournament:** enable all five approved candidates on independent accounts. Gate: operational stability and frozen manifest.

**Phase 4 — Evaluation:** compare common-period results, costs, uncertainty, and incident effects. Deliver ranked report with no-promotion option.

**Phase 5 — Confirmation:** continue unchanged versions over later unseen data. Deliver promotion/continue/reject recommendation.

**Phase 6 — Optional future live-readiness:** separate BRD/change approval; no automatic transition.

No delivery dates are committed. Broker provisioning, data availability, and approved strategy specificity determine estimates.

Phase 1 replay uses Alpaca options history from February 2024. Phase 2 fits within the three paper accounts one Alpaca login provides. Phase 3 cannot start until D02 is resolved. [v1.1]

## 18. Risks and mitigations

**False winner from selection:** retain variant history and separate confirmation data.

**Unrealistic paper fills:** dual ledgers, conservative scenarios, and execution sensitivity reports.

**Shared software defect:** independent fixtures, reconciliation, and parity tests.

**Unfair comparison:** common manifest and explicit exceptions.

**Mid-test tuning:** immutable versions and approval gate.

**Agent hallucination/overreach:** read-only permissions and calculation-backed reports.

**Broker account limitations:** verify five-account provisioning early; internal simulated portfolios require an explicit scope change.

**Data outage/rate limits:** bounded subscriptions, health gates, and incident-aware evaluation.

**Look-ahead bias:** event-time feature construction and replay tests.

**Excess platform complexity:** modular monolith, incremental rollout, and reuse.

**All strategies unprofitable:** record valid rejection; do not force a winner or enable live trading.

**Alpaca account limit:** Alpaca allows up to three paper accounts per login. Ask Alpaca in writing whether a second login is permitted; two logins would cover the five candidate accounts. If not, run in stages or approve an internal-ledger scope change. [v1.2]

**Provider feed incidents:** Alpaca's OPRA feed delivered stale quotes for several sessions after an August 7 deployment. Freshness gates (MD-04), stale-feed handling (AC-08), and incident-aware evaluation (FR-10) cover this. [v1.1]

**Single host:** the Mac mini is a single point of failure. UPS, auto-restart, off-machine backups, and startup reconciliation mitigate it. **[v1.1]**

**Small account constrains contracts:** at $10,000 equity, a small per-trade budget may allow only further out-of-the-money SPY contracts with wider relative spreads and less reliable paper fills. Review budget-feasible deltas before fixing D07. **[v1.1]**

**Agent plan limits:** Claude Pro usage caps can delay agent reports; trading is unaffected (AG-09). Upgrade path: Max plan or API. **[v1.1]**

**Channels feature maturity:** the Telegram channel is a research preview and may change; SSH and Remote Control remain available. **[v1.1]**

## 19. Open-decision register

v1.1 adds a status for each decision. **[v1.1]**

| ID | Decision | Status (v1.1) | Notes |
|---|---|---|---|
| D01 | Identify existing strategy, code quality, account mode, history, and whether it maps to a named candidate. | Resolved (v1.2) | No incumbent strategy. Nothing is carried forward; Phase 0 removed. |
| D02 | Confirm broker, availability of five isolated paper accounts, and any optional sixth benchmark. | Open | Alpaca selected. Five accounts needed, no sixth benchmark (v1.2). Three paper accounts per login; second login pending Alpaca's confirmation. Reviewed and not preferred: tastytrade sandbox (resets daily, synthetic fills), TradeStation SIM (instant fills), Tradier paper (15-minute delayed data), IBKR (one paper account per live account). |
| D03 | Select market-data provider, option coverage, entitlements, and quote/Greek requirements. | Proposed | Alpaca Algo Trader Plus. ThetaData Options Value as optional deeper history. |
| D04 | Approve eligible underlyings and subscription limits. | Resolved | SPY only. |
| D05 | Approve all strategy thresholds and indicator definitions. | Open | |
| D06 | Approve expiration/delta/liquidity policy and cutoff behavior. | Open | Defined as a function of the per-trade risk budget (CS-01A); depends on D07. |
| D07 | Approve starting equity, sizing basis, account limits, and session-loss handling. | Partially resolved | Starting equity $10,000 per account. Per-trade risk and daily loss limit TBD. |
| D08 | Approve order types, replacement limits, timeouts, and failure contingencies. | Open | |
| D09 | Approve evaluation/confirmation periods, evidence criteria, and statistical decision policy. | Open | Replay history from February 2024 proposed for Phase 1. |
| D10 | Approve cost model, marking conventions, infrastructure-cost allocation, and outage exclusions. | Open | |
| D11 | Approve deployment, authentication, backup, retention, and measurable service objectives. | Partially resolved | Dedicated Mac mini, native macOS + Supervisor, SSH access, one repo, after-close changes. Backup targets, retention, and RPO/RTO open. |
| D12 | Approve dashboard stack and agent model/provider, tool permissions, privacy, and budget. | Partially resolved | Budget under $100/month; Claude Pro available; Telegram preferred. Streamlit and Claude Code proposed. Tool permissions open. |

## 20. Traceability summary

**Concurrent comparison:** OBJ-01 -> FR-01 through FR-05 -> architecture/strategy workers -> AC-01, AC-02, AC-19.

**Credible profitability:** OBJ-02 -> evaluation/dual-ledger requirements -> analytics -> AC-11, AC-12, AC-17.

**Operational diagnosis:** OBJ-03 -> MD/OM/reconciliation requirements -> event store -> AC-05 through AC-09.

**Reproducibility:** OBJ-04 -> versioning/manifests -> experiment manager -> AC-03, AC-10.

**Confirmation:** OBJ-05 -> evaluation phases/promotion gates -> reports -> no automatic promotion.

**Agent assistance:** OBJ-06 -> AG-01 through AG-10 -> isolated supervisor -> AC-13, AC-14.

**No forced winner:** OBJ-07 -> FR-12 -> dashboard/promotion logic -> AC-12.

**Release integrity:** OBJ-04 -> NFR-15, Section 9.3 -> startup preflight -> AC-21. **[v1.1]**

**Budget-driven sizing:** OBJ-02 -> CS-01A, CS-05, RK-07 -> account runtime -> AC-22. **[v1.1]**

**Agent isolation:** OBJ-06 -> AG-11 to AG-13, NFR-14 -> separate OS user -> AC-24. **[v1.1]**

## 21. Approval and implementation readiness

This BRD is a proposed implementation baseline, not authorization to deploy trading or a claim of profitability. Business approval covers scope, experiment mode, and primary objective. Research approval covers exact strategy rules, metrics, and decision policy. Operational approval covers accounts, credentials, risk, recovery, and incident controls.

Implementation may begin on common interfaces and replay infrastructure while open decisions are resolved. Paper execution must not begin until account bindings, strategy specifications, risk limits, order policies, and startup/recovery checks are approved.

Before paper execution, D02 (account count) and D07 (per-trade risk and daily loss limit) must also be resolved. **[v1.1]**

The final outcome is a reproducible, independently accounted comparison with evidence strong enough to support a next research decision—even if that decision is to reject every strategy.